

# EDA Report: Geldium Dataset

Subject: Delinquency Risk Analysis

Date: April 2, 2026

## 1. Key Insights & Data Quality Observations

### Key Findings

- Missing Values:** Significant gaps identified in *Income*, *Credit Utilization*, and *Payment History*.
- Inconsistencies:** Invalid or unrealistic entries found within the *Payment History* field.
- Outliers:** *Credit Utilization* levels were detected exceeding 100%, indicating potential data entry errors or extreme financial distress.
- Early Risk Indicators:** High credit utilization, missed payments, and low income levels are the primary drivers of delinquency risk.

### Summary

The Geldium dataset contains several data quality issues, including missing values and inconsistencies in key financial fields. Credit utilization and payment history appear to be the strongest predictors of delinquency risk. Because extreme values and outliers were identified, rigorous preprocessing is required to ensure accurate and reliable predictive modeling.

## 2. Missing Data Handling Strategy

| Issue                      | Method            | Justification                                                  |
|----------------------------|-------------------|----------------------------------------------------------------|
| Missing Income             | Median Imputation | Minimizes the influence of extreme income outliers.            |
| Missing Credit Utilization | Mean Imputation   | Preserves the overall distribution of the data.                |
| Missing Payment History    | Drop Rows         | This is a critical feature; synthetic data may bias the model. |

## 3. Risk Indicators & Patterns

### Primary High-Risk Indicators

- High Credit Utilization:** Indicates an over-reliance on credit and a higher probability of default.
- Missed Payment History:** The strongest historical predictor of future delinquency.
- Low Income:** Suggests a limited capacity for debt repayment.

4. **High Debt-to-Income (DTI) Ratio:** A clear sign of existing financial stress.

### Behavioral Insights

Analysis reveals that customers with high credit utilization and a history of missed payments have a significantly higher likelihood of delinquency. Notably, high-income individuals are not exempt from risk; some exhibit "lifestyle creep" or overspending behaviors that result in high delinquency risk despite their earnings.

## 4. Final Conclusion

The initial EDA of the Geldium dataset highlights critical data quality issues that must be mitigated. Key variables — specifically **Credit Utilization** and **Payment History** — are the cornerstones of identifying delinquency risk. By implementing the suggested missing data treatments and focusing on the identified risk patterns, the resulting predictive model will achieve higher precision and reliability.